



Project proposal: FairLoan

Names: Divine Uwase Ingabire

Monia Nijimbere

Lyse Claudia Irera

Serge Ishimwe

Cohort B

CSIS Department, Ashesi University

CS 254: Introduction to Artificial Intelligence,

Dr. Daniel Addo

A. Overview

Project Name: FairLoan

The problem: Lenders must decide whether to approve a loan before granting it, and these decisions have a measurable financial impact on the banking sector. Consumer debt data from early 2026 shows an overall loan delinquency rate of 4.76% and a credit card delinquency rate of 2.92% (Federal Reserve Board of Governors, DRCCLACBS series), showing that many households continue to struggle with debt repayment. Since defaults cause financial losses, the approval stage is the key point where banks manage risk. However, limited review time can lead to rushed or inconsistent decisions. Our project addresses this challenge by predicting loan approval outcomes from historical application data, giving loan officers a consistent, data-driven reference to support their decisions.

Project details: FairLoan is a machine learning system that supports loan approval decisions using historical application data. It trains a supervised classification model to predict whether a loan application is likely to be approved based on features such as income, credit history, employment status, and loan details. The project also evaluates the model's accuracy and conducts a fairness audit to assess whether predictions are biased on features such as gender, race, etc, thereby promoting fair and explainable lending decisions.

The specific user: The loan officer or credit analyst who reviews applications is the person most directly affected by this problem. Since this review currently depends heavily on individual experience and interpretation, two reviewers can look at the same file and reach different conclusions, leading to inconsistent results in a process that is supposed to be objective. Our model gives the loan officer a consistent, data-backed reference point they can weigh alongside their own expertise, instead of replacing their judgment entirely.

Stakeholders:

1. The bank assumes the financial risk of default, so its lending process must be both profitable and defensible to regulators and shareholders.
2. The applicant's access to credit depends directly on the outcome, and this is significant given that roughly 1.3 billion adults worldwide still lack a bank account, with half of them concentrated in just eight economies.
3. Regulators and compliance bodies, since lending decisions must be explainable and free from unfair bias. This is why an explainability layer in this kind of model is imperative for accountability.

AI technique(s) used

We will use supervised classification because our dataset contains labeled historical outcomes for each applicant, and the dataset's target variable is a binary categorical outcome (approved/rejected), not a continuous value.

Dataset: We will mainly use Kaggle to source datasets. Our main dataset is the Loan Approval Classification dataset, which contains 45 000 records and 14 variables. It can be found here:

<https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data>

Tools: For this project, we will use Python along with the pandas and numpy libraries for data handling, including cleaning, transforming, and preparing the loan applicant dataset for analysis. For building our classification models, we will use Scikit-learn, which is suited to our binary classification task of predicting loan approval or rejection. Development will take place in Google Colab for an interactive environment for data exploration, model building, and iterative testing.

B. Team roles

Names	Roles
Divine Uwase Ingabire	Coordinator(scheduling/logistics)
Serge Ishimwe	Team leader
Lyse Claudia Irera	Quality control manager
Monia Nijimbere	Researcher

Our team will work collaboratively on all technical aspects of the project, data preparation, modeling, and evaluation, and each member will have a non-technical role (Researcher, Coordinator, quality control manager, Team leader) to ensure the project stays organized, well-documented, and clearly communicated.

C. Milestone plan

Timeline	Tasks	Delivarables
Week 9 Jul 13 - 17	- Choose project - Submit project proposal - Finalize dataset & confirm target column	Project proposal

Week 10 Jul 20–24	• Clean data & handle missing values • Encode categorical features • Set up GitHub repo with README skeleton	None
Week 11 Jul 27–31	• Build baseline model (logistic regression) • Build the full pipeline working end-to-end • Commit code	Working baseline model (Progress check)
Week 12 Aug 3 –7	• Build and tune the second model • Begin evaluation • Prepare & deliver in-lab progress demo	Progress Demo (Thu Aug 6, ungraded, feedback only)
Week 13 Aug 10–14	• Run a fairness audit • Finalize evaluation & select final model • Finalize report, README, and slides	None
Week 14 Aug 17–18	• Submit final report + GitHub link • Deliver an oral presentation	Final Report (PDF) + GitHub link; Oral Presentation

References

12 CFR Part 1002 - Equal Credit Opportunity Act (Regulation B) | Consumer Financial Protection Bureau. (n.d.). Consumer Financial Protection Bureau.

<https://www.consumerfinance.gov/rules-policy/regulations/1002/>

Board of Governors of the Federal Reserve System (US). (2026). Delinquency Rate on Credit Card Loans, All Commercial Banks [DRCCLACBS]. In *FRED*. er: Federal Reserve Bank of St. Louis. Retrieved July 16, 2026, from <https://fred.stlouisfed.org/series/DRCCLACBS>

The Global Findex database. (2020, February 4). Data | FinDev Gateway.

<https://www.findevgateway.org/data/global-findex-database>

UCI Machine Learning Repository. (n.d.).

<https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>